



The GOES-R Satellite Series: NOAA's Next-Generation Environmental Watch

Remote Sensing & Satellite Tech

February 2026

A collaborative NASA-NOAA program that represents the most significant technological advancement in geostationary observations in the last 30 years, providing high-definition monitoring of Earth and the Sun.



Kaushal Singh
CH22BTECH11018

The Four-Satellite Constellation



Satellites renamed from letters to numbers upon reaching geostationary orbit

GOES-16

Launched: 2016

On-orbit Backup
105° W

GOES-17

Launched: 2018

On-orbit Backup
Storage

GOES-18

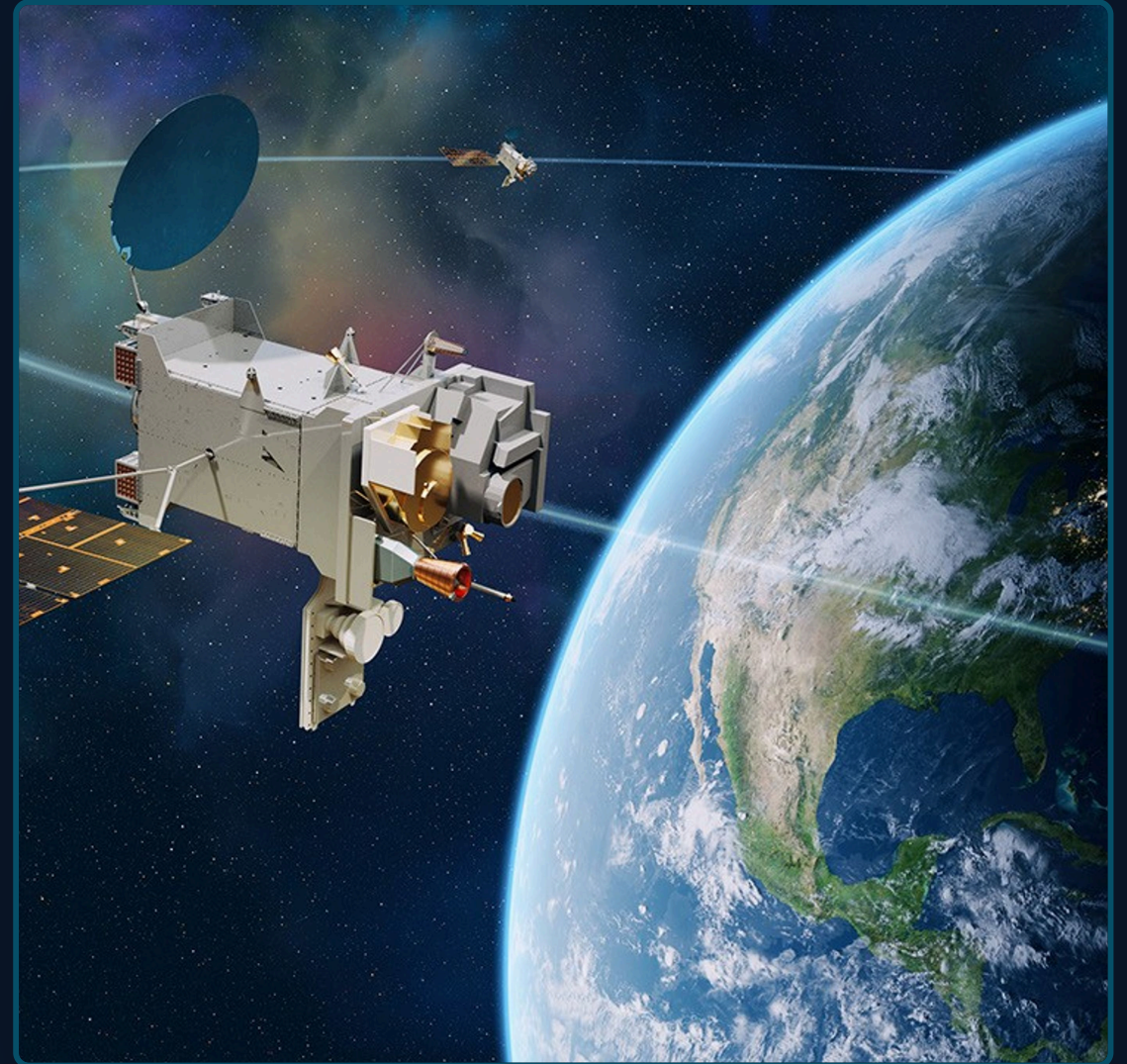
Launched: 2022

GOES-West Operational
137.2° W

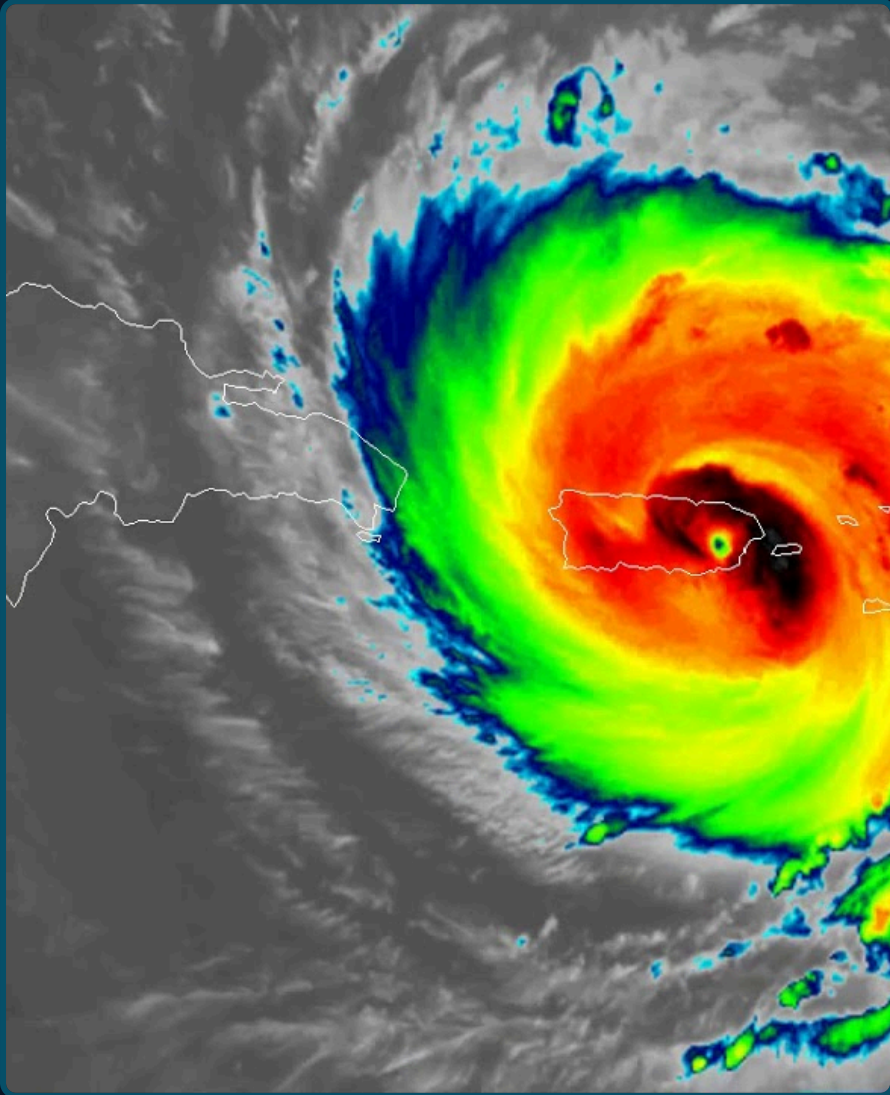
GOES-19

Launched: 2024

GOES-East Operational
75.2° W



Constant Vigilance for Public Safety



Severe Weather Monitoring

Improved accuracy & lead time for hurricane, tornado, and flood warnings



Space Weather Tracking

Monitoring solar activity to protect power grids, satellites, and GPS systems



Total Lightning Detection

Real-time mapping of in-cloud and cloud-to-ground lightning for rapid storm identification



Environmental Stewardship

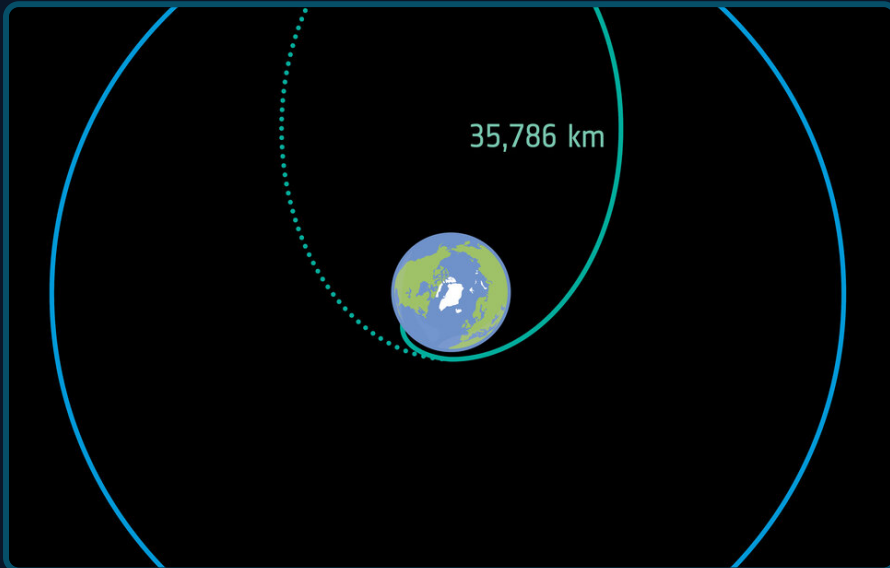
Tracking air quality, smoke, volcanic ash, and sea surface temperatures

Technical Excellence at 35,786 km

Geostationary Orbit (GEO)

Altitude: **35,786 km**

Satellites rotate with Earth, remaining fixed over the same point



Advanced Baseline Imager (ABI)

16 spectral bands (visible & IR)

Resolution: 0.5 km - 2 km



Geostationary Lightning Mapper (GLM)

Real-time lightning detector • First in GEO



Space Weather Suite

SUVI

Solar UV Imager

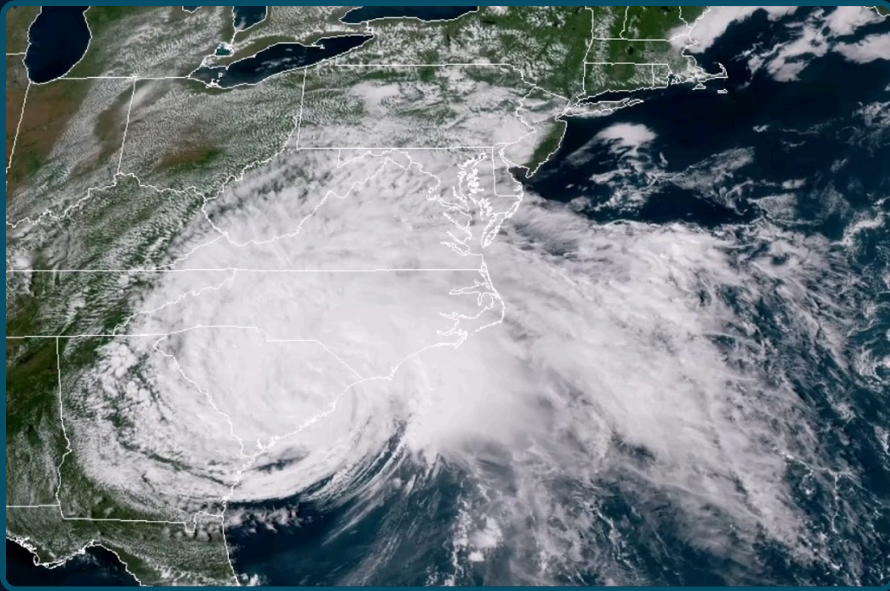
EXIS

Solar Irradiance

CCOR

Compact Coronagraph
(GOES-19)

From Hurricanes to Wildfires



Meteorology

Real-time tracking of hurricane centers & intensifying storm cores via GLM lightning jumps



Aviation Safety

Improved detection of low-level fog & volcanic ash plumes for efficient flight routing



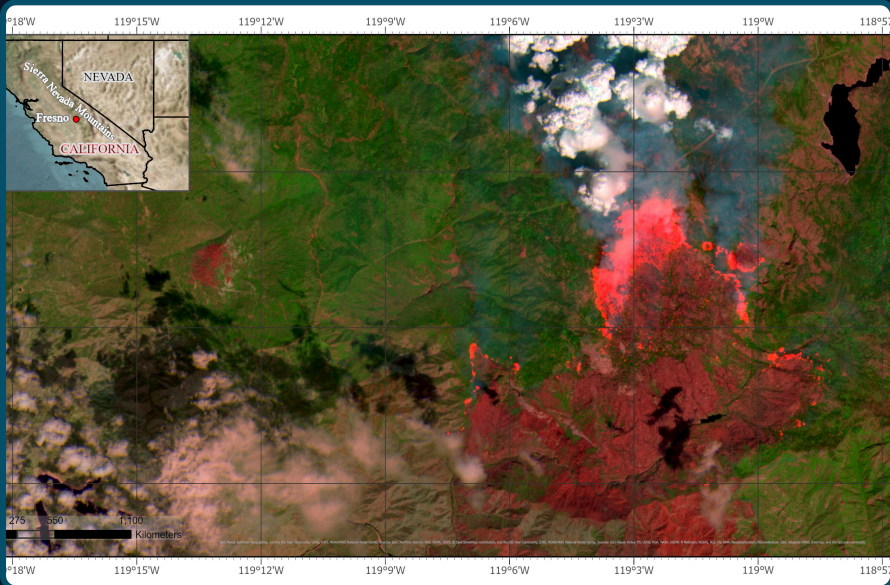
Disaster Response

Identifying "hot spots" for wildfires before ground reports, enabling rapid response



Climate Science

Long-term monitoring of ocean temperatures & atmospheric aerosols for climate variability studies



Challenges and Transitions



Physical Constraints

Poor resolution at poles due to Earth's curvature • Polar-orbiting satellites (JPSS) fill gaps



Data Volume Challenges

Massive ground-station infrastructure and advanced AI required for data processing



Technical Issues

GOES-17 faced cooling system anomalies limiting nighttime infrared imaging



Future Replacement

Operational through **2036** • Will be replaced by **GeoXO** program with higher resolution and new sensors



The GOES-R Series: Protecting Our Future

A vital infrastructure for modern weather forecasting



Thank You for Your Attention